Tribhuvan University

Faculty of Humanities and Social Sciences

TRAVEL_X

A PROJECT REPORT

Submitted to

Department of Computer Application

Swastik College

Chardobato-Bhaktapur

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted By:

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Under the supervision of

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SUPERVISOR'S RECOMMENDATION

It is our pleasure to recommend the project titled "Travel_X", a smart transportation booking system completed by Kriti Basnet and Subash Shrestha under our supervision, for submission and final evaluation. This work has been carried out as a part of the requirements for the Bachelor of Computer Application degree.

……..

Susan Sunuwar

SUPERVISOR

Bachelor of Computer Application

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LETTER OF APPROVAL

This is to certify that this project prepared by Kriti Basnet and Subash Shrestha entitled "**Travel_X"** in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion, it is satisfactory in the scope and quality as a project for the required degree.

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ABSTRACT

The Travel_X Transportation Management and Booking System is a modern, user friendly web application designed to simplify vehicle search, booking, and management for both passengers and transport service providers. Developed using HTML, CSS, JavaScript, and SQL Server, this system integrates essential features to ensure a smooth and efficient experience. The platform supports secure user registration and login, with different access roles for customers, vehicle owners, and administrators. Users can browse various vehicle categories such as cars, vans, buses, trucks, and two/three-wheelers check availability, and place booking requests easily. A smart filtering system helps users find vehicles based on type, capacity, location, and rental duration. One of the system's key features is its organized rate and schedule management, allowing customers to compare options and choose the most suitable transport service. The interface is clean, responsive, and visually appealing, offering seamless navigation across devices. On the backend, the system ensures data security and consistency using SQL Server, while authentication and booking operations are handled using structured logic in JavaScript and server-side processing. The system also supports features such as booking history, vehicle management, and image integration for better user interaction.

Keywords: *Transportation Booking System, Vehicle Management, Multi-Vehicle Booking Platform, Online Transport Services.*

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Thank you to everyone who contributed in any way your support truly made this project possible.

Sincerely,

Kriti Basnet

Subash Shrestha

TABLE OF CONTENTS

ABSTRACT i
ACKNOWLEDGEMENT ii
TABLE OF CONTENTS iii
LIST OF ABBREVIATIONS v
LIST OF FIGURES vi
LIST OF TABLES vii
CHAPTER 1: INTRODUCTION 1
1.1 Introduction 1
1.2 Problem Statement 1
1.3 Objectives 2
1.4 Scope and Limitation 2
1.4.1 Scope 2
1.4.2 Limitation 2
1.5 Development Methodology 2
1.6 Report Organization 4
CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW 5
2.1 Background Study 5
2.2 Literature Review 6
CHAPTER 3: SYSTEM ANALYSIS AND DESIGN 7
3.1 System Analysis 7
3.1.1 Requirement Analysis 7
3.1.2 Feasibility Analysis 8
3.1.3 Data Modeling (ER-Diagram) 10
3.1.4 Process Modelling (DFD) 11
3.2 System Design 12
3.2.1 Architectural Design 12
3.2.2 Database Schema Design 13
3.2.3 Interface Design 13
3.3 Algorithm Details 14
CHAPTER 4: IMPLEMENTATION AND TESTING 16
4.1 Implementation 16
4.1.1 Tools Used 16

4.1.2 Implementation details of modules 17
4.2 Testing 19
CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATION 25
5.1 Conclusion 25
5.2 Lesson Learnt/ Outcome 25
5.3 Future Recommendation 26
APPENDICES 27
REFERENCES 30

LIST OF ABBREVIATIONS

PHP Hypertext Preprocessor

SQL Structured Query Language

HTML Hyper Text Markup Language

CSS Cascading Style Sheets

JS JavaScript

DB Database

API Application Programming Interface

LIST OF FIGURES

Figure 1.1: Waterfall model of Travel_X 3
Figure 3.2:Use case diagram of Travel_X 7
Figure 3.3:Gantt Chart of Travel_X. 10
Figure 3.4: ER of Travel_X 10
Figure 3.5: 0-Level DFD of Travel_X 11
Figure 3.6: 1-Level DFD of Travel_X 11
Figure 3.7: 2-level DFD of Travel_X 12
Figure 3.8: Architectural Design of Travel_X 12
Figure 3.9: Database Schema Design of Travel_X 13

LIST OF TABLES

Table 4.1: Unit Test for User Login. 20
Table 4.2: Unit Test for Travel Search Module 21
Table 4.3: Unit Test for Booking Module 22
Table 4.4: Unit Test for User Profile Module 23
Table 4.5: Unit Test for Admin Management Module 24

CHAPTER 1: INTRODUCTION

1.1 Introduction

Travel_X is a transportation and vehicle-booking platform designed to help users find and reserve the most suitable vehicles based on their travel needs, budget, destination, and availability. Whether someone needs a car, van, bus, bike, or three-wheeler, Travel_X simplifies the entire search and booking process by delivering clear options and a smooth user experience.

With the growing variety of travel modes and rental services, selecting the right vehicle can often feel confusing and time-consuming. Travel_X addresses this challenge by organizing vehicle information in a structured way, allowing users to quickly browse available options, compare features, and choose the one that best fits their purpose—be it daily travel, tours, events, or long-distance trips.

The main goal of Travel_X is to make vehicle booking easier, faster, and more reliable. The system provides categorized vehicle listings, filtering options, and a clean user interface that improves decision-making. By offering well-organized information and an efficient reservation workflow, Travel_X ensures that users can confidently select the right vehicle without unnecessary hassle.

1.2 Problem Statement

The traditional method of finding and booking vehicles often lacks flexibility, provides limited filtering options, and does not always reflect real-time availability. This can make users confused, cause delays, and lead to an overall frustrating experience. As booking platforms expand, ensuring data security, system reliability, and scalability becomes even more important.

Travel_X addresses these challenges by offering a more intelligent and efficient vehiclebooking solution. The system allows users to filter vehicles by type, capacity, purpose, and budget, while ensuring up-to-date availability information. With a clean and responsive interface, secure booking processes, and well-structured vehicle listings, Travel_X aims to deliver a smooth, reliable, and user-friendly experience for all types of travelers.

1.3 Objectives

The primary objectives of the Travel_X Vehicle Booking System are as follows:

i. To develop a smart recommendation system that suggests suitable vehicles based on user preferences such as price, vehicle type, capacity, and travel requirements.
ii. To create a scalable system that can accommodate various vehicle categories and expand with future requirements.

1.4 Scope and Limitation

1.4.1 Scope

i. Users can browse and book a wide range of vehicles, including cars, vans, buses, and other multi-wheel transport options based on their travel needs.
ii. Vehicle owners or service providers can register and list their vehicles**,** update details, set pricing, and manage availability.
iii. The system recommends suitable vehicles using a content-based filtering approach that compares price, category, seating capacity, and user preference patterns.

1.4.2 Limitation

i. Real-time availability may depend on timely updates by vehicle owners**,** and any delay in updating availability may lead to inconsistencies during booking.
ii. The system requires a stable internet connection, as all booking operations, searches, and updates are handled online.
iii. Vehicle variety may be limited by the number of owners who register**,** which can affect the range of recommendations in certain regions.

1.5 Development Methodology

The Travel_X system is developed using the Waterfall Model. The system follows a structured sequence of phases, starting with planning**,** analysis**,** design**,** and implementation**.** In the Waterfall Model, the output of each phase serves as the input for the next phase, ensuring a systematic and organized approach to developing the Travel_X platform. This methodology helps in maintaining clarity, reducing errors, and ensuring that each stage of development aligns with the objectives of Travel_X.

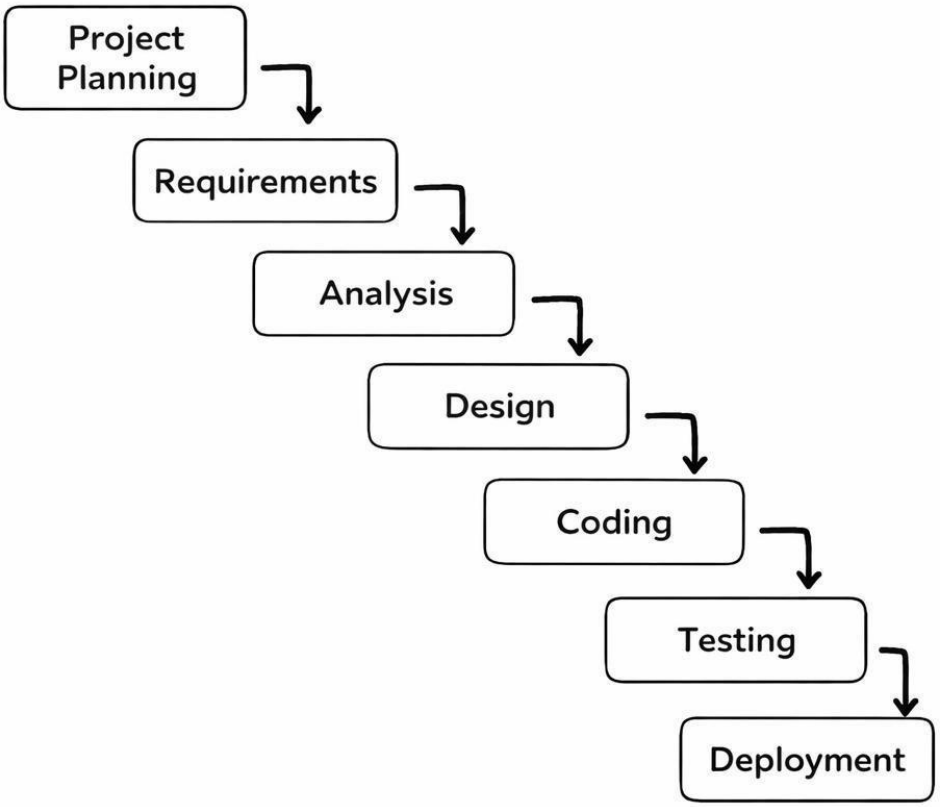


Figure 1.1: Waterfall model of Travel_X

The Waterfall Model was chosen for developing the Travel_X system because all project goals and requirements were clearly defined at the beginning. Since no changes were expected during development, a linear and structured approach was suitable. All requirements were first collected and analyzed in detail. Based on this analysis, the system design, user interfaces, and algorithms were planned. After completing the design phase, the system was implemented through coding. Each phase was completed before moving to the next, with no need to revisit earlier stages. This ensured a smooth development process, consistent quality, and a final system that matched the original project plan.

1.6 Report Organization

This report provides comprehensive information about the Travel_X project. The report is organized as follows:

Chapter 1: This chapter begins with an introduction to Travel_X and its purpose, followed by the problem statements in existing travel systems. It also outlines the objectives, scope, and limitations of the Travel_X project..

Chapter 2: This chapter provides a brief analysis of the existing travel systems and highlights their limitations.

Chapter 3: This chapter covers the system analysis for Travel_X, including the project's feasibility, data modeling, and process modeling. It also details the techniques used to determine system requirements. Following this, the system design is presented, including database schema design, interface design, process design, and user interface design.

Chapter 4: This chapter explains the tools and technologies used in the Travel_X project, covering both the front-end and back-end along with their purposes. It also describes the project modules and the testing processes, including unit testing and system testing..

Chapter 5: The final chapter presents the conclusion**,** summarizing how Travel_X was developed, the key findings, and overall outcomes. It also includes recommendations and suggestions for future enhancements of the Travel_X system.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Traditional vehicle rental services in Nepal have long been dependent on manual processes, physical visits to rental offices, and phone-based inquiries. This approach was timeconsuming, limited in choice, and often required local knowledge to navigate the complex transportation landscape of Nepal's diverse terrain. Early online platforms provided basic listings but lacked real-time availability, personalized recommendations, and comprehensive service integration.

The emergence of digital platforms revolutionized the vehicle rental industry globally, with companies like Enterprise, Hertz, and local platforms in Nepal beginning to offer online booking capabilities. However, these systems initially provided limited functionality—basic vehicle listings, static pricing, and minimal personalization. Users faced challenges in finding vehicles suited to Nepal's unique road conditions, varying altitude requirements, and diverse travel needs.

Travel_X represents the next evolution in Nepal's vehicle rental ecosystem. Unlike earlier systems that offered generic listings, Travel_X incorporates intelligent algorithms that consider multiple factors specific to the Nepali context:

i. Terrain-specific vehicle recommendations (mountain-friendly SUVs vs. city-appropriate sedans)
ii. Altitude performance requirements for Himalayan destinations
iii. Seasonal variations affecting road conditions
iv. Cultural and festival-based travel patterns
v. Multi-lingual support for both domestic and international travelers

Modern web technologies have enabled Travel_X to provide real-time vehicle availability, dynamic pricing based on demand and seasonality, and integrated insurance and driver services. The system leverages frameworks like React for responsive user interfaces and Node.js for scalable backend processing, allowing for seamless booking experiences across devices.

2.2 Literature Review

User interface design significantly impacts vehicle rental platform success by enhancing usability, building trust, and facilitating decision-making. Features such as intuitive vehicle categorization, transparent pricing structures with all-inclusive costs, and responsive design that works across devices are critical for user satisfaction in the Nepali market. [1] Studies show that clear display of vehicle specifications (ground clearance, fuel type, transmission) and route suitability information can reduce booking abandonment rates by up to 40% in transportation platforms.

Key factors influencing vehicle rental decisions in Nepal include vehicle type suitability for specific terrains, rental pricing transparency, driver availability, insurance coverage, and customer reviews. Real-time availability tracking and transparent pricing that includes fuel policies, insurance, and additional charges significantly enhance user confidence. [2] Research indicates that users in developing markets particularly value clear information about vehicle condition, safety features, and local driver qualifications when making rental decisions.

In the context of transportation recommendation systems, Ricci conducted a comprehensive review of intelligent mobility solutions, categorizing vehicle recommendation approaches into: content-based methods focusing on vehicle attributes and trip requirements; collaborative filtering leveraging similar users' rental patterns; and knowledge-based systems incorporating domain expertise about terrain, weather, and local regulations. Their research demonstrates that hybrid recommendation systems, combining multiple approaches, achieve significantly better performance in matching travelers with appropriate vehicles, especially in complex environments like Nepal's diverse geography. [3]

Current vehicle rental platforms in Nepal face several limitations: many rely on basic search filters that don't adequately address terrain-specific requirements; pricing often lacks transparency with hidden charges; mobile experiences are frequently non-responsive; and recommendation systems fail to account for local factors such as seasonal road conditions, altitude considerations, and festival-related travel patterns. Additionally, most systems suffer from the cold-start problem where new users receive generic recommendations due to insufficient data.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

3.1.1 Requirement Analysis

1. Functional Requirement

i. **User Registration and Authentication**: The system allows Customers, Vehicle Owners, and Admins to register and securely log in using role-based access to ensure authorized use of features.

ii. **Vehicle Listing and Management**: Vehicle Owners can add, update, and manage vehicle details such as type, price, availability, and specifications, which are displayed to customers in real time.

iii. **Search, Booking, and Reservation Management**: Customers can search vehicles using filters, view details, make bookings, modify reservations, and select optional services based on availability.

iv. **Payment and Transaction Processing**: The system supports secure payment processing, booking confirmation, transaction tracking, and payment history for both customers and vehicle owners.

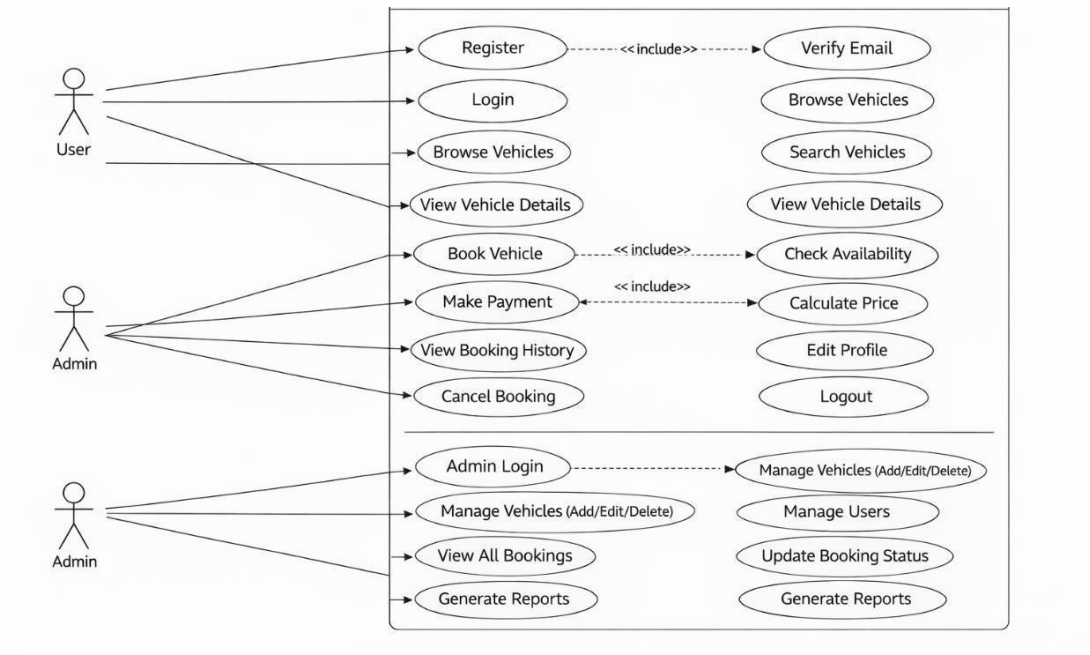


Figure 3.2:Use case diagram of Travel_X

i) Non-Functional Requirement

a. **Accessibility**

The Travel_X system is a web-based application accessible anytime and from anywhere with an internet connection. Users across Nepal can access the platform easily without installing additional software.

b. **Usability**

The system follows standard UX/UI design principles to ensure simplicity and ease of use. Clear navigation, user-friendly interfaces, and intuitive booking steps enhance the overall user experience.

c. **Performance**

The platform is optimized for fast loading and smooth operation, even under Nepal's varying internet bandwidth conditions. Efficient database queries and caching mechanisms help improve response time.

d. **Security**

Strong security measures are implemented to protect user data and online transactions. Features like encrypted passwords, secure payment gateways, and authentication mechanisms ensure data privacy.

e. **Reliability**

The system maintains consistent uptime and stable performance. Regular backups and error-handling mechanisms ensure continuous service without data loss.

3.1.2 Feasibility Analysis

The feasibility study for Travel_X is conducted to evaluate whether the proposed system is viable by analyzing technical, operational, and economic factors. This study helps determine the practicality and effectiveness of developing and implementing the Travel_X platform.

i.Technical Feasibility

The Travel_X system is developed using PHP as the backend programming language and MySQL as the database management system. The frontend is designed using HTML, CSS, and JavaScript to ensure a responsive and interactive user interface. The system is developed and tested using XAMPP, which provides an integrated environment with Apache and MySQL support. All the technologies and tools required to develop and run

this system are widely available, open-source, and well-supported. Therefore, the Travel_X system is considered technically feasible.

ii. Operational Feasibility

The Travel_X system is operationally feasible as it simplifies the process of vehicle searching and booking for users. Customers can browse available vehicles, check pricing, and make reservations online without visiting a physical office. The system is easy to use and does not require advanced technical knowledge, making it accessible to a wide range of users. This reduces manual effort, saves time, and improves service efficiency. Hence, the system is operationally feasible.

iii. Economical Feasibility

The Travel_X system is economically feasible since it is developed using free and opensource software such as PHP, MySQL, and XAMPP. The main costs involved are server hosting, internet services, and manpower for system maintenance. Revenue can be generated through vehicle booking commissions, premium services, and promotional advertisements. As the benefits outweigh the development and operational costs, the system is economically feasible.

iv. Schedule Feasibility

Schedule feasibility focuses on whether the system can be completed within the planned timeframe. The development of the Travel_X system was properly planned and executed in phases such as requirement analysis, design, development, testing, and deployment. Each phase was completed within the allocated time, ensuring that the project met its deadline successfully. Therefore, the system is feasible in terms of schedule.

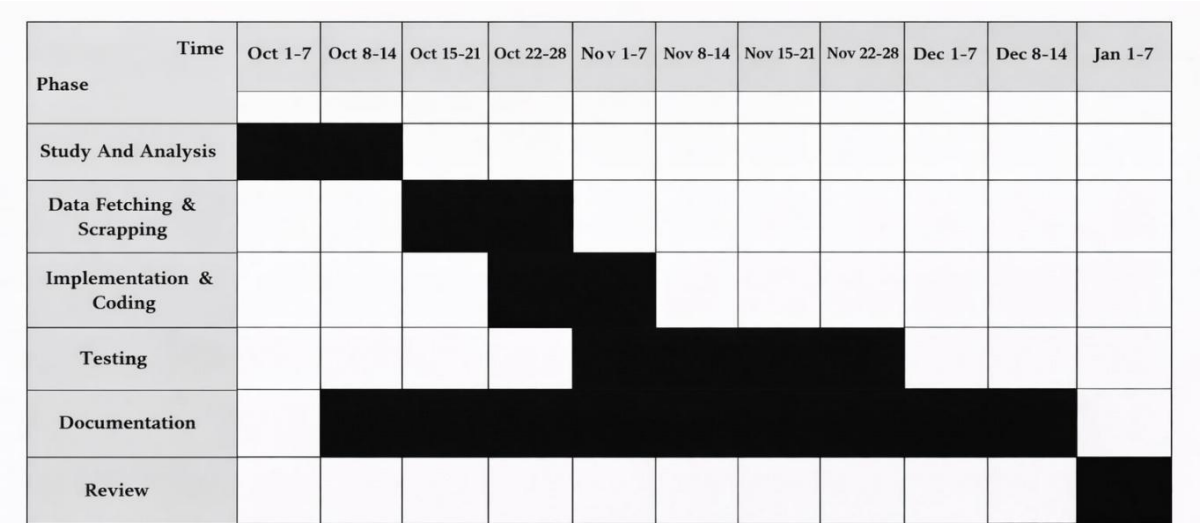


Figure 3.3:Gantt Chart of Travel_X

3.1.3 Data Modeling (ER-Diagram)

In the proposed Entity–Relationship (ER) Diagram for the Travel_X system, the database is structured around three core entities: User, Vehicle, and Booking. Each entity consists of a set of attributes that store essential information required for the operation of the system. The relationships between these entities are clearly defined with appropriate cardinalities to ensure efficient data organization and integrity.

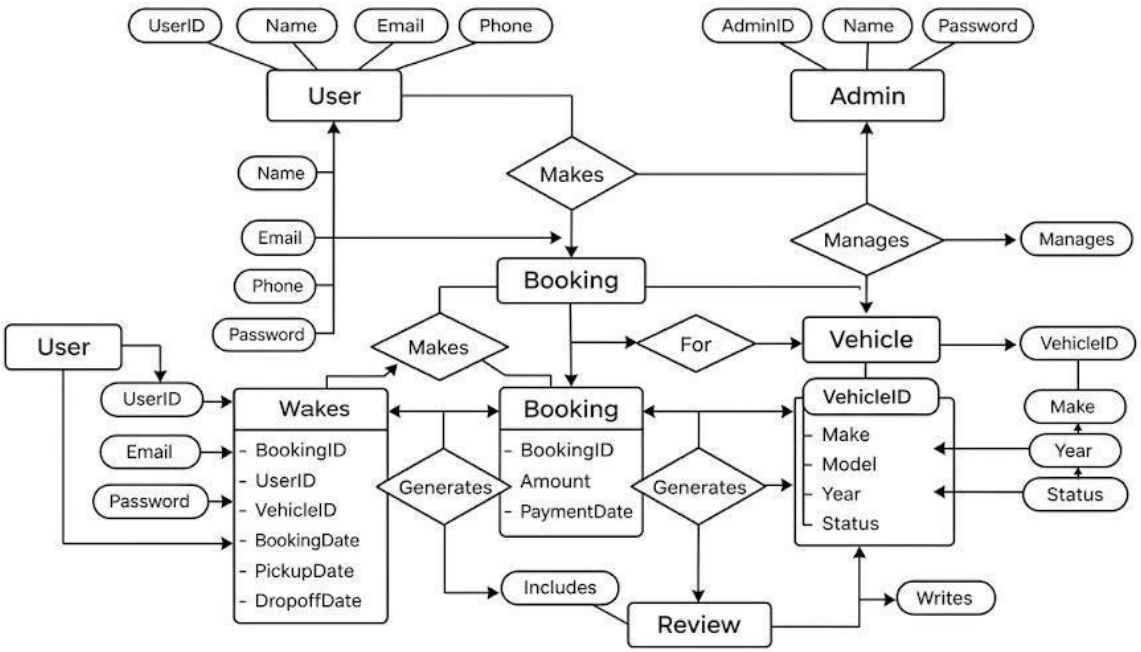


Figure 3.4: ER of Travel_X

3.1.4 Process Modelling (DFD)

The Data Flow Diagram (DFD) of the Travel_X Transportation Management System is developed up to Level 1. These diagrams illustrate how data flows within the system and how different processes interact with users, vehicles, and the database to ensure smooth operation of the platform.

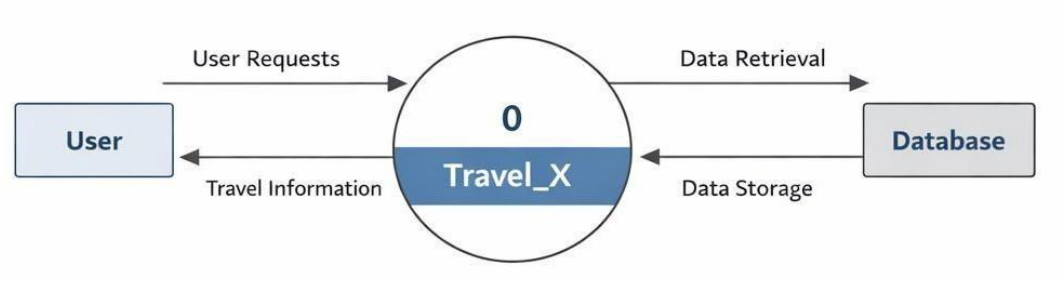


Figure 3.5: 0-Level DFD of Travel_X

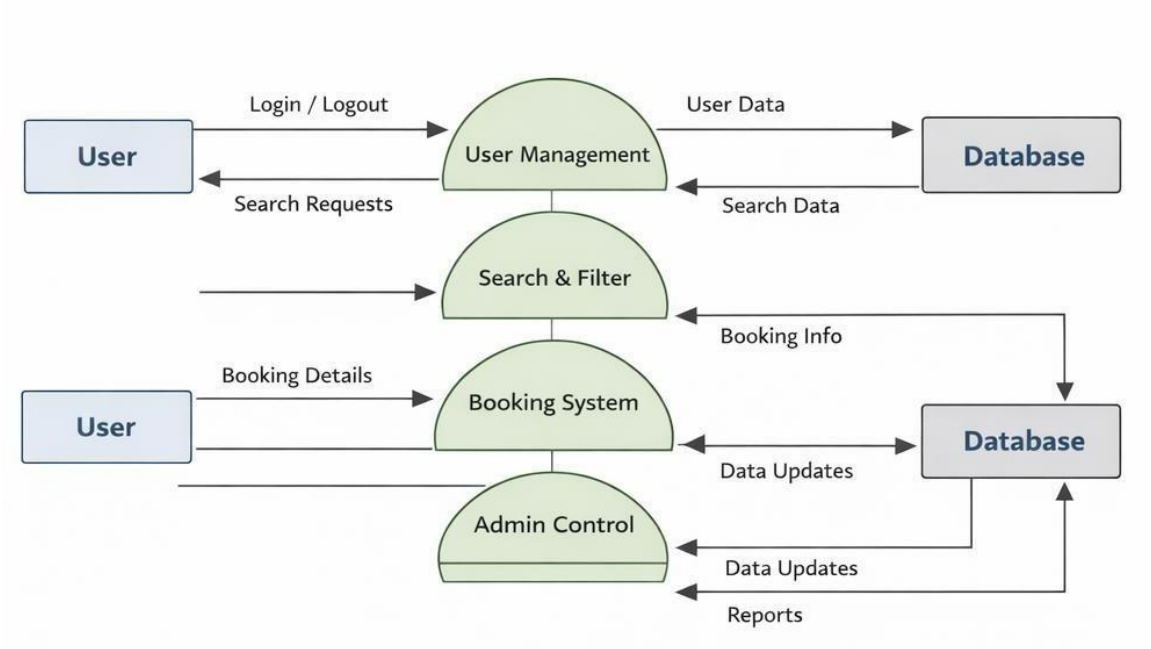


Figure 3.6: 1-Level DFD of Travel_X

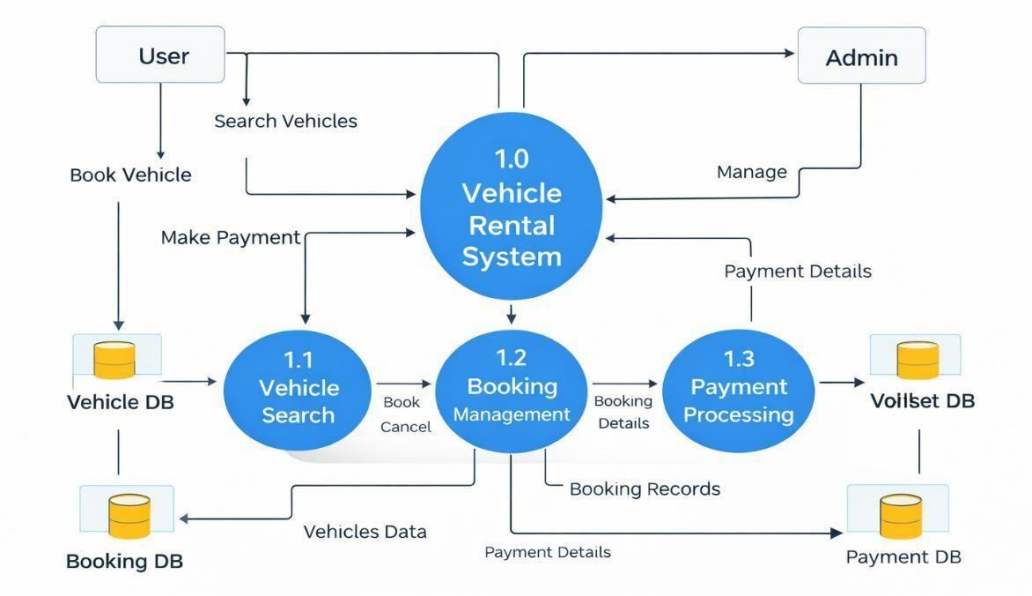


Figure 3.7: 2-level DFD of Travel_X

3.2 System Design

To implement the various functional requirements of the Travel_X Transportation Management System in a visual and structured manner, several system design diagrams have been developed. These diagrams provide a clear representation of how different components of the system interact and operate together.

3.2.1 Architectural Design

The Travel_X Transportation Management System follows a three-tier architecture**,** which separates the system into three distinct layers: the User Interface Layer, the Application (Web Server) Layer, and the Database Layer.

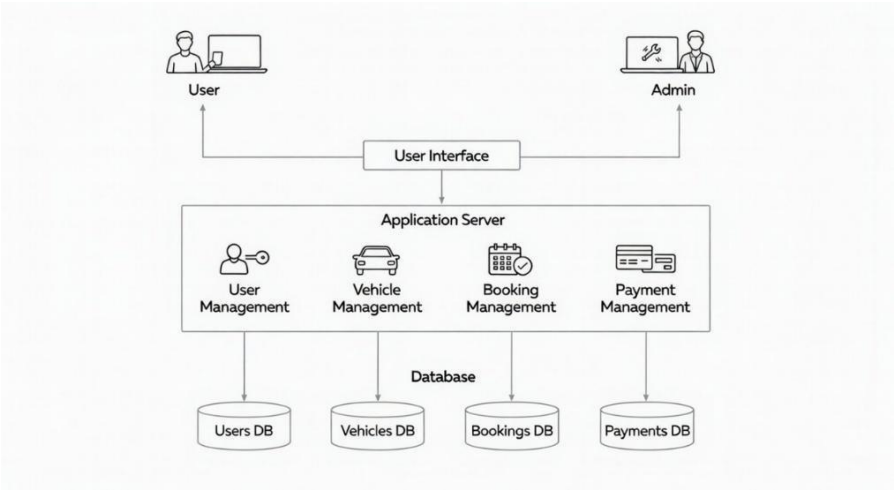


Figure 3.8: Architectural Design of Travel_X

3.2.2 Database Schema Design

The database schema design of Travel_X presents an overview of the database structure by describing the tables, their attributes, and the relationships among them. It helps in understanding how data is stored, managed, and connected within the system. The database schema diagram of the system is as follow:

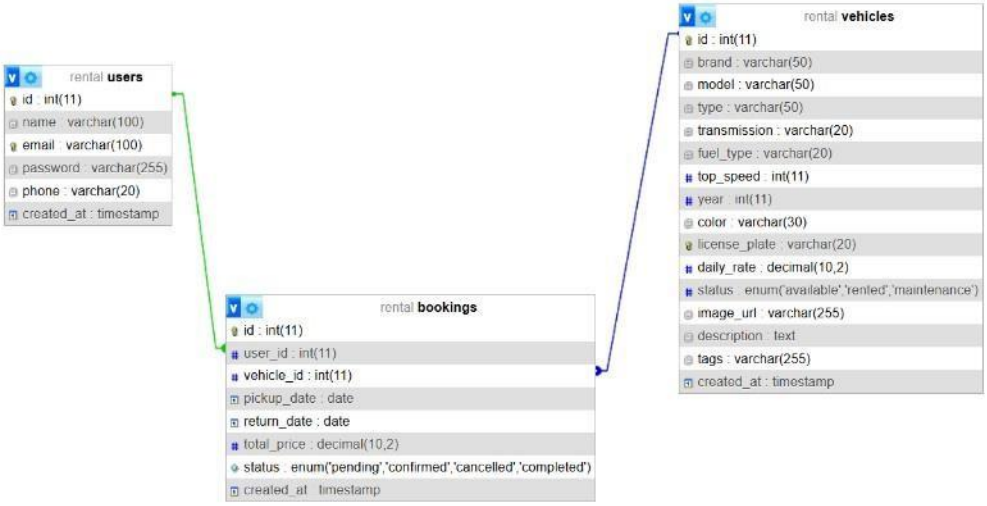
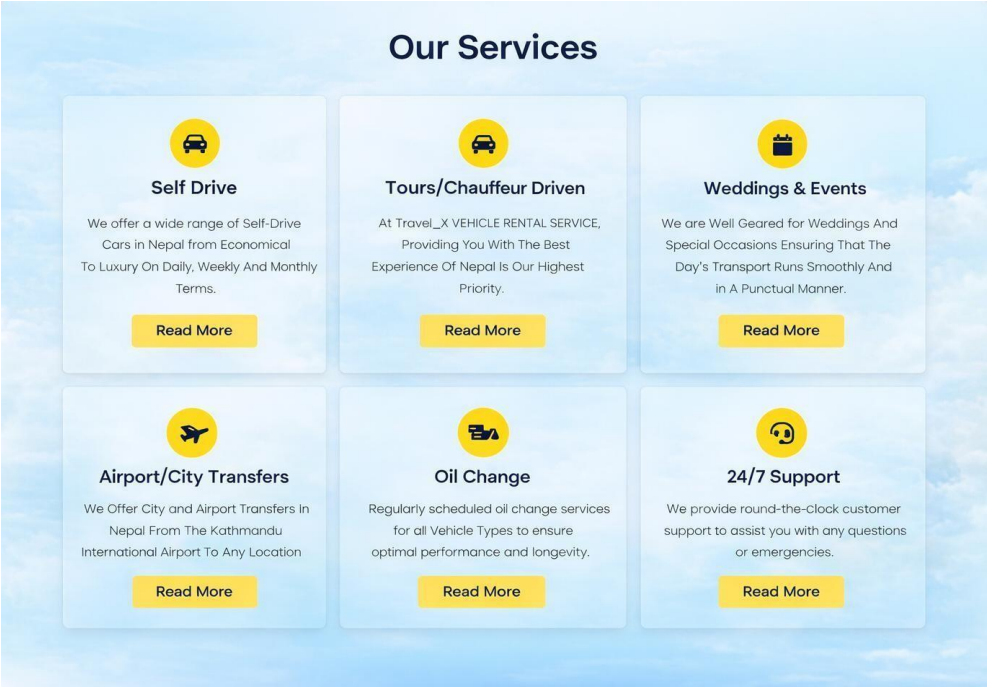


Figure 3.9: Database Schema Design of Travel_X

3.2.3 Interface Design

The user interface of Travel_X is designed to collect necessary inputs from users and display the results in a clear and user-friendly manner. The UI design of the system is as follow:



3.3 Algorithm Details

The Travel_X system uses a searching algorithm and content-based algorithm to help users quickly find suitable transportation options based on their requirements. When a user enters search criteria such as vehicle type, destination, travel date, or service category, the system scans the available vehicle records stored in the database.

3.3.1 Searching Algorithm

Steps:

1. Accept User Input
2. All Vehicles
3. Filter Matching Vehicles
4. Check Availability
5. Sort Results
6. Display Results

A vehicle is considered a match if:

VehicleMatch = (VehicleType == UserType)
AND (Availability == TRUE)

AND (ServiceCategory == UserRequirement)

Only vehicles that satisfy all conditions are included in the search results.

This searching-based approach ensures fast retrieval**,** accurate results, and efficient vehicle selection, making Travel_X reliable for transportation booking and logistics services.

3.3.2. Content-Based Recommendation Algorithm

The content-based recommendation algorithm suggests vehicles by comparing user preferences with vehicle features.

Steps:

1. Collect user preferences (price range, vehicle type, capacity, travel purpose)
2. Extract attributes of available vehicles from the database
3. Compare user preferences with vehicle attributes
4. Calculate similarity score
5. Rank vehicles based on similarity
6. Recommend top matching vehicles

Recommendation Formula:

RecommendationScore=w1*(VehicleTypeMatch)+w2*(PriceMatch)+ w3*(Capacity Match) + w4*(Service Match)

Where:

- w1, w2, w3, w4 are weights assigned to each feature

Matching Logic:

A vehicle is recommended if:

SimilarityScore $\geq$ Threshold Value

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

In this phase of Travel_X**,** the actual coding of different modules and functions takes place. Each component of the system is developed according to the specifications and design outlined in the system design phase.

4.1.1 Tools Used

i) **Front-End Tools:**

a. CSS

CSS is used in Travel_X to design and enhance the look of the web pages. It helps in creating a clean and organized layout, making the interface more user-friendly and visually consistent across the application.

b. JavaScript

JavaScript adds interactivity to Travel_X**,** enabling features such as dynamic content updates, form validations, and responsive elements. It ensures that users can interact smoothly with the application without unnecessary page reloads, improving overall usability. ii) **Back-End Tools**

a. PHP:

PHP is used on the server side to communicate with the database, process user requests, and deliver dynamic content to the frontend. This combination ensures smooth interaction between the database and the user interface, maintaining data consistency and efficient system performance.

iii) **Documentation Tools**

a. Draw.io

Draw.io, a free web-based platform, is used in Travel_X to create diagrams for system design and planning. Its drag-and-drop interface allows for easy creation of flowcharts, architecture diagrams, and other visual representations, simplifying the design process. b. Microsoft Word

Microsoft Word is used for preparing all project-related documentation for Travel_X**,** including reports, descriptions of system modules, and analysis. It provides a structured way to present information clearly and professionally.

4.1.2 Implementation details of modules

i. **Recommendation Module**

The recommendation module in Travel_X is based on a **Content-Based Filtering Algorithm**, which suggests suitable vehicles by comparing user preferences with vehicle attributes such as type, price, capacity, and service category. Users provide details such as source, destination, and travel date. The system processes this data and retrieves matching travel records from the database.

This module applies a **content-based filtering approach**, where the system analyzes user input (such as source, destination, date, vehicle type, and preferences) and matches it with stored vehicle attributes. PHP is used to process user requests, and SQL queries retrieve matching records based on similarity between user requirements and vehicle features.

Code Implementation:

```
<?php include("connection.php");
if(isset($_POST['search'])) {
$source = $_POST['source'];
$destination = $_POST['destination'];
$date = $_POST['travel_date'];
$query = "SELECT * FROM travels
WHERE source='$source'
AND destination='$destination'
AND travel_date='$date'";
$result = mysqli_query($conn, $query);
if(mysqli_num_rows($result) > 0) { while($row
= mysqli_fetch_assoc($result)) { echo "Vehicle:
"
$row['vehicle_type']."<br>";          echo
"Price: ".$row['price']."<br><br>";
}
} else
 {    echo    "No    travel    options
available.";
```

```
}
}
?>
```

The content-based algorithm ensures that users receive personalized and relevant vehicle suggestions without relying on other users' data, making it efficient and suitable for systems with limited user history.

ii. **Login Module**

The Login Module in Travel_X manages user authentication by validating user credentials.
It verifies the entered email and password against stored records in the database.
Only authenticated users are allowed to access protected features of the system.

The module is implemented using PHP sessions and SQL queries to ensure secure login functionality.
Code Implementation:

```
<?php
session_start();
include("connection.php");
if(isset($_POST['login'])) {
$email = $_POST['email'];
$password = $_POST['password'];
$query = "SELECT * FROM users
WHERE email='$email'
AND password='$password'"; $result =
mysqli_query($conn, $query);
if(mysqli_num_rows($result) == 1) {
$_SESSION['user_email'] = $email;
header("Location: dashboard.php");
} else  { echo "Invalid email or
password";
";
}
}
```

?>

This code validates user credentials and creates a session upon successful login. If authentication fails, an error message is displayed.

4.2 Testing

Testing is the process of identifying and fixing errors in a system. It is essential to ensure the quality, reliability, and performance of the application. Travel_X undergoes testing using various test cases to verify that all features work as intended. For the Travel_X system, comprehensive testing is performed using well-defined test cases to validate that all functionalities operate correctly and efficiently. This process helps in detecting bugs, improving user experience, and ensuring system stability. The different types of testing conducted for the system are as follows:

4.2.1 Test Case for Unit Testing

Unit testing is conducted on individual modules of the Travel_X system to ensure that each component functions correctly and independently. This process helps in identifying errors at an early stage before the modules are integrated with other parts of the system.

i) User Authentication

Table 4.1: Unit Test for User Login

ID	Description	Steps	Expected Result
1.1	Valid Login Credentials	1. Open the Travel_X login page 2. Enter valid email and password 3. Click the "Sign In" button	User is successfully logged in and redirected to the dashboard
1.2	Invalid Login Credentials	1. Open the login page 2. Enter invalid email and password 3. Click on the "Sign In" button	An error message is displayed indicating invalid login credentials
1.3	Empty Login Fields	1. Open the login page 2. Leave email and password fields empty 3. Click the "Sign In" button	Validation error messages are displayed for required fields
1.4	Invalid Emai Format	1. Open the login page 2. Enter invalid email format and password 3. Click on the "Sign In" button	Error message displayed for invalid email format
1.5	User Logout	1. User is logged in 2. Click on "Sign Out" button 3. Confirm logout	User is successfully logged out and redirected to home page

Table 4.2: Unit Test for Travel Search Module

ID	Description	Steps	Expected Result
2.1	Valid Search Input	1. Open the search page 2. Enter valid travel details (destination, dates) 3. Click “Search”	List of available travel options is displayed correctly.
2.2	Invalid Search Input	1. Open the search page 2. Enter invalid or incomplete travel details 3. Click “Search”	Error message or no results message is displayed
2.3	Empty Search Fields	1. Open the search page 2. Leave all search fields empty 3. Click “Search”	Validation messages prompt user to enter required fields
2.4	Filter Options	1. Perform a search 2. Apply filters (price, type, rating) 3. Click “Apply”	Search results are updated according to the applied filters

ii) Booking Management

Table 4.3: Unit Test for Booking Module

ID	Description	Steps	Expected Result
3.1	Successful Booking	1. Select a travel option 2. Enter valid booking details 3. Click "Confirm Booking"	Booking is successfully saved and confirmation displayed
3.2	Invalid Booking Details	1. Select a travel option 2. Enter invalid or incomplete booking details 3. Click "Confirm Booking"	Error messages are displayed indicating invalid input
3.3	Cancel Booking	1. Open user bookings 2. Select a booking 3. Click "Cancel" and confirm	Booking is canceled and updated in the system
3.4	Payment Validation	1. Enter payment information 2. Submit payment 3. Verify payment	Payment is processed successfully or error displayed for invalid details

Table 4.4: Unit Test for User Profile Module

ID	Description	Steps	Expected Result
4.1	Update Profile Details	1. Open user profile 2. Modify name, email, or other details 3. Click "Save"	Profile is updated successfully
4.2	Invalid Profile Input	1. Open user profile 2. Enter invalid data (e.g., wrong email format) 3. Click "Save"	Error message displayed for invalid input
4.3	Change Password	1. Open change password section 2. Enter old password and new password 3. Click "Update"	Password is updated successfully
4.4	Empty Profile Fields	1. Open user profile 2. Leave mandatory fields empty 3. Click "Save"	Validation messages prompt user to fill required fields

iii) Admin Management

Table 4.5: Unit Test for Admin Management Module

ID	Description	Steps	Expected Result
5.1	View All Users	1. Navigate to the Admin panel 2. Click on “User Management” 3. View the user list	All registered users of the Travel_X system are displayed with complete details
5.2	View All Travel Service	1. Navigate to the Admin panel s2. Click on “Service Management” 3. View the service list	All travel services are displayed with details
5.3	Delete Travel Services	1. Navigate to the Admin panel 2. Click on “Details” 3. Select a service from the list 4. Click “Delete Service” 5. Confirm deletion	The selected travel service is successfully deleted from the Travel_X system.
5.4	Delete User	1. Navigate to Admin panel 2. Click ”Details” 3. Select user from list 4. Click “Delete User” 5. Confirm deletion	The selected user is successfully deleted from the Travel_X system

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATION

5.1 Conclusion

The Travel_X system successfully delivers relevant and timely travel and hotel recommendations using a simple yet effective search filtering algorithm. By focusing on realtime data and user context, the system enhances the overall travel planning and booking experience without relying on complex machine learning models. The seamless integration of backend recommendation logic with a clean, responsive, and user-friendly frontend ensures that users receive meaningful and accurate suggestions efficiently.

This balanced approach emphasizes performance, simplicity, and usability, making Travel_X suitable for real-world deployment. Overall, the Travel_X platform adds significant value by helping users quickly discover suitable travel options, improving user engagement, satisfaction, and decision-making efficiency..

5.2 Lesson Learnt/ Outcome

During the development of Travel_X**,** the author gained practical experience in designing and implementing efficient searching algorithms suitable for real-world applications. The system uses a price-based searching and filtering approach to retrieve relevant travel and hotel options based on user input. This demonstrated how simple and well-structured searching techniques can provide fast and accurate results without relying on complex machine learning models.

This project highlighted the importance of real-time data searching, efficient filtering, and sorting mechanisms to improve system performance and user experience. The author also learned to handle edge cases effectively by implementing fallback search results when exact matches were not available. Additionally, the project provided hands-on experience in optimizing search queries, integrating backend search logic with frontend interfaces, and designing a clean and user-friendly UI. Overall, the development of Travel_X strengthened the author’s understanding of searching algorithms, full-stack development, and usercentered system design.

5.3 Future Recommendation

Although Travel_X is fully functional and suitable for real-world use, there is scope for further enhancement. Future improvements may include: Incorporate machine learning techniques to personalize recommendations based on user behavior and preferences.

i. Improving the searching algorithm by incorporating advanced search techniques that consider multiple parameters such as price range, location, facilities, ratings, and availability to provide more accurate search results.
ii. Implementing optimized and faster search mechanisms**,** such as indexed searching and query optimization, to improve system performance when handling large volumes of travel data.

APPENDICES

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WELCOME TO Travel_X "The Name You Can Trust"
VEHICLE RENTAL SERVICE IN NEPAL
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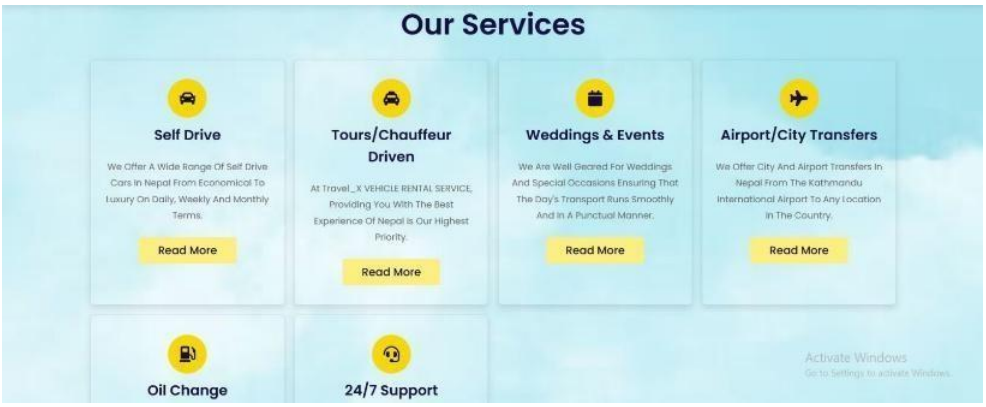
Rent Vehicles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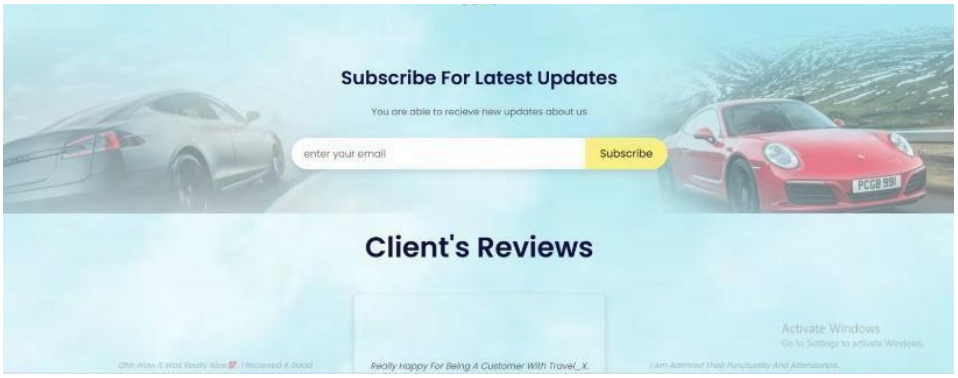
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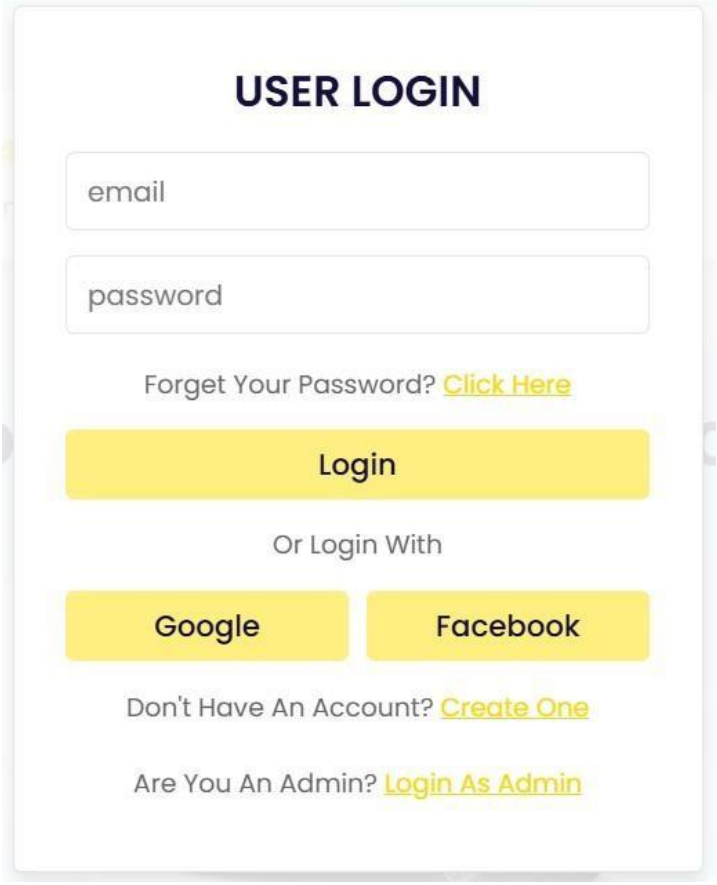
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720+
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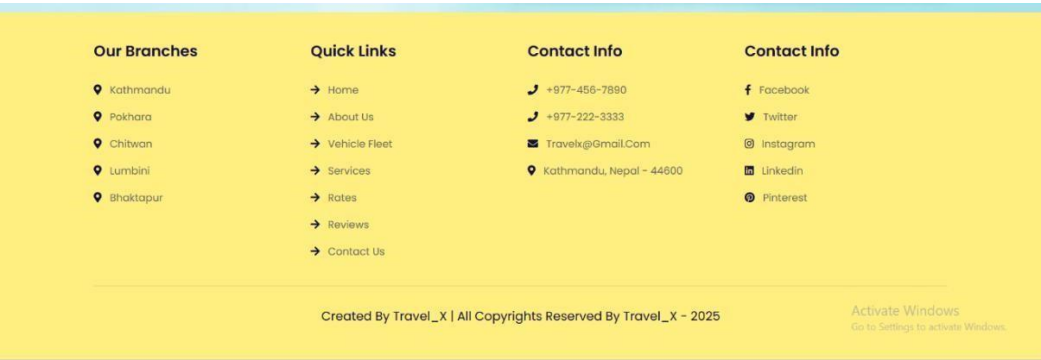
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